0.1 Counting Elements of Disjoint Sets: The Addition Rule

The Addition Rule

Suppose a finite set A equals the union of k distinct mutually disjoint subsets $A_1, A_2, \ldots, A_k$ Then:

$$N(A) = N(A_1) + N(A_2) + \cdots + N(A_k).$$

The Difference Rule

If A is a finite set and B is a subset of A, then:

$$N(A - B) = N(A) - N(B).$$

Formula for the probability of the complement of an Event

If S is a finite sample space and A is an event in S, then:

$$P(A^c) = 1 - P(A)$$

The Inclusion/Exclusion Rule for Two or Three Sets

If A, B and C are any finite sets, then

$$N(A \cup B) = N(A) + N(B) - B(A \cap B)$$

and

$$N(A \cup B \cup C) = N(A)+N(B)+N(C)-N(A \cap B)-N(A \cup C)-N(B \cap C)+N(A \cap B \cap C).$$

Pigeonhole Principle

A function from one finite set to a smaller finite set cannot be one-to-one: There must be at least two elements in the domain that have the same image in the co-domain.

Generalized Pigeonhole Principle

For any function f from a finite set X with n elements to a finite set Y with m elements and for any positive integer k, if $k < n/m$, there there is some $y \in Y$ such that y is the image of at least k + *1* distinct elements of X

0.1.1 Generalized Pigeonhole Principle (Contrapositive Form)

For any function f from a finite set X with n elements to a finite set Y with m elements and for any positive integer k, if for each $y \in Y, f^1(y)$ has at most k elements, then X has at most km elements; in other words, $n \leq km$.

0.1.2 The Pigeonhole Principle

For any function f from a finite set X with n elements to a finite set Y with m elements, if $n > m$, then f is not one-to-one.

0.1.3 One-to-One and Onto for Finite Sets

Let X and Y be finite sets with the same number of elements and suppose f is a function from X to Y. Then f is one-to-one if, and only if, f is onto.